

Dylan Ong

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EDUCATION

University of Wisconsin-Madison | Madison, WI

May 2026

Bachelor of Science, Double Major: **Computer Science & Data Science**

GPA: 3.93/4.0

- Certificate (Minor): **Accounting Fundamentals**, Honors: Dean's List for all 6 semesters
- Relevant Coursework: Data Structures & Algorithms, Data Science Modeling, Artificial Intelligence, Machine Learning, Big Data, Database Management, Operating Systems, Finance, Risk Management

WORK EXPERIENCE

CS Capstone Software Engineer | UW-Madison Department of Neuroscience

January 2026 - May 2026

- Designed and built **multithreaded architecture** for real-time eye-tracking experiments using **Python**
- Integrated application with Pupil Labs **eye tracker API** to stream and process gaze data in real time
- Developed interactive GUI using **Kivy** to manage experiment flow
- Implemented **data pipeline** and **SQLite** database for storing session data and experiment results

Senior Research Intern for Development | Center for Research on Complex Thinking

October 2024 - May 2026

- **Promoted to Senior Intern** to **manage a team of developers** building and maintaining educational research platforms (Epistemic Network Analysis, iPlan, Codey).
- Led **quality assurance testing** and **designed workflow processes** that improved team efficiency.
- Conducted **data analysis** in **R** and **provided technical support** for **30+** institutional researchers.

PROJECTS

Strava Analyzer | *Fullstack Development* | Python, Scikit-learn, Streamlit, REST APIs, OAuth

View project on: <https://strava-analyzers.streamlit.app/>

- Developed a **machine learning pipeline** in **Python** to predict running pace using **Random Forest regression**
- Built a **Streamlit web app** enabling dynamic input of planned runs and **visualization** of past performance trends
- Implemented **Strava API integration**, including **OAuth** token refresh, to automate activity retrieval

Financial Stock Analysis | *Fullstack Development* | Python, TensorFlow/Keras, scikit-learn, Flask, React, REST APIs

View project on: <https://financial-stock-analysis.onrender.com/>

- Built a multi-stock **LSTM model** in **Python** to predict stock prices using historical data and scaling techniques
- Built a **Flask backend** to serve data and real-time model predictions using **REST APIs**
- Integrated **React frontend** components to visualize price trends and prediction outputs
- Implemented **data preprocessing** and **inference pipelines** for time-series financial data

MiniSpark Project | *Operating Systems Project* | C, Data Processing, Concurrency

- Built a simplified **Spark-like data processing engine** using **RDD** abstractions in **C**
- Implemented **multithreaded** task execution with synchronization via **locks** and **condition variables**
- Designed **concurrent workflows** for parallel data processing

SKILLS

- **Programming:** Java, C/C++, Python, R, SQL
- **App Development:** React, Node.js, JavaScript, HTML/CSS
- **Data Platforms & Distributed Systems:** Kafka, Cassandra, Spark, Hadoop, BigQuery
- **Frameworks & Tools:** Git, Docker, Bash, Unix, CI/CD, REST APIs, Visual Studio
- **Languages:** Bahasa Indonesia (native), English (fluent), Mandarin Chinese (conversational)

CERTIFICATIONS

Completion of Apple Foundation Program | Apple Developer Academy | Surabaya, Indonesia

June - July 2019

Certificate of Chinese Proficiency (HSK 4) | Xin Zhong Course School | Surabaya, Indonesia

July 2010 - May 2021